Erwan MARTIN

*Available from **July 2026** for an **Artificial Intelligence Engineer** position*

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Experience

R&D Data Scientist, Fixed-term contract, TotalEnergies – Pau, France 🇫🇷 — July 2025 – July 2026, 1 year

*Project: Develop an **AI-based marine biodiversity monitoring solution** using Distributed Acoustic Sensing (DAS) data.*

- Built a database of acoustic signatures of marine species from scientific publications
- Processed **53 TB** of images on an HPC cluster (**Slurm** & **LSF** schedulers): applied **Butterworth/F–K filtering**, energy computation, and **FFT**-based spectrogram extraction with **Python**, parallelized via **Dask** (**–89%** processing time)
- Trained convolutional autoencoders (**CNN**) for spectrogram **feature extraction** (**Tensorflow**)
- Performed **unsupervised clustering** (**K-Means, HDBSCAN**) on embeddings for species identification
- Conducted spatio-temporal species localization via **geometric optimization** on coordinates (x, y, z)

Data Scientist / Android Developer Intern, Aneo – Boulogne-Billancourt, France 🇫🇷 — May 2024 – November 2024, 6 months

*Project: Finalize **an Android application for hepatic spectrometry** to estimate macrosteatosis from patient data, in collaboration with the **BOPA Innovation Chair** (AP-HP, Institut Mines-Télécom, Université Paris-Saclay).*

- Completed the **Android spectrometry app** (**React, Ionic, TypeScript**)
- Implemented a **CI/CD pipeline** (**Python**) and **cloud storage** (**AWS S3, Lambda**)
- **Deployed** the application for patient data collection at Paul Brousse Hospital, working closely with surgical teams
- Developed and monitored ML models via **Kedro** and **MLflow** (**regression, Random Forest, SVM**)

Data Scientist Intern, Dow – Tarragona, Spain 🇪🇸 — Sept. 2023 – March 2024, 6 months

*Project: Develop an ML model to **predict the mechanical properties of plastic films** based on their chemical composition.*

- **Extracted**, **cleaned**, and **integrated** experimental data (**SQL, Python, Pandas, NumPy**)
- Explored **supervised models**: **Deep Learning**, **XGBoost** (**Tensorflow**, **scikit-learn**)
- **Deployed** two **XGBoost** models (R^2 = **0.94** and **0.82**) on a **Streamlit** app for industrial use

Data Scientist Intern, THALES – Vélizy-Villacoublay, France 🇫🇷 — May 2022 – September 2022, 4 months

Project: Develop an ML model to predict the trajectory of an electron beam from gun parameters.

- Performed **Monte Carlo simulations** (**ANSYS: Design Modeler & Meshing**, **Persalys**) for data generation
- Designed **AutoML** and **Deep Learning** models (**H2O**, **TensorFlow**)

Education

CY TECH (ex E.I.S.T.I), Engineering degree (Master's level) in **Applied Mathematics**, specialization in **Data Science / Artificial Intelligence** — September 2020 – Nov. 2024

- **Coursework:** Machine Learning • Deep Learning • Probability • Statistics • Time Series • Graph Theory • Optimization • Parallel Computing • Signal Processing • NLP • Constraint Programming

Lycée Jean-Baptiste Corot, Preparatory classes for engineering schools (CPGE) – Mathematics and Physics track (MPSI – MP) — September 2018 – July 2020

Skills

Languages: 🇫🇷 French (Native, C2), 🇬🇧 English (Advanced, C1, TOEIC: 910/990), 🇩🇪 German (Beginner, A2)
PyTools: Numpy • Pandas • iGraph • Scikit-Learn • Tensorflow • Pytorch • NLTK • Librosa • Selenium • Kedro • MLflow
Programming Languages: Python (mainly) • R • SQL • TypeScript
Databases: MySQL • MariaDB • AWS S3
Technologies: Git • Azure DevOps • Anaconda • Jupyter Notebooks • Visual Studio Code • OracleDB • RStudio • Zotero
Data Visualization: Streamlit • Plotly • Matplotlib • Seaborn • SAS • Power BI • Dash
Operating Systems: Linux, Windows
Work Methodology: Agile Scrum